

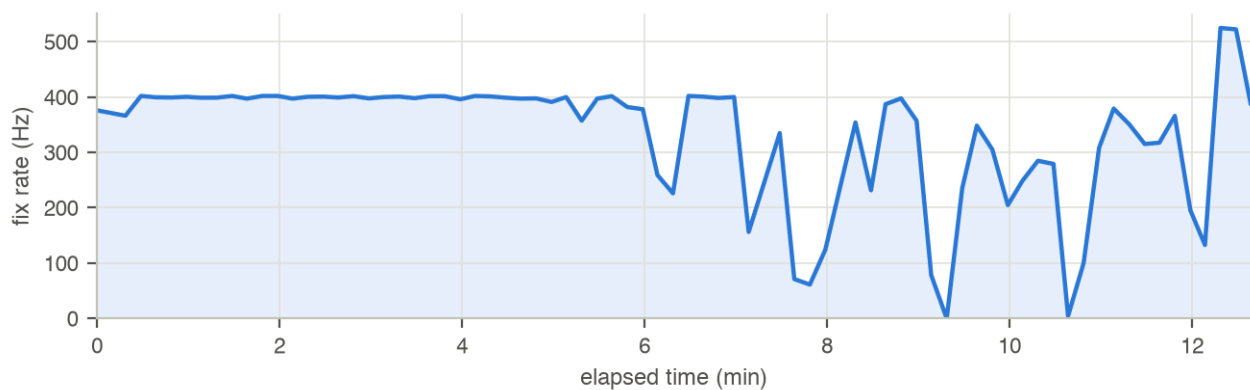
Radio range test analysis

Source: data/base_station_rate_stats_2026-09-27.csv - a radio-congestion bench run walked/driven from the base out to roughly 2 miles. The base's own LOG_RATE_STATS logger samples fix rate and sync-error rate every 10 seconds.

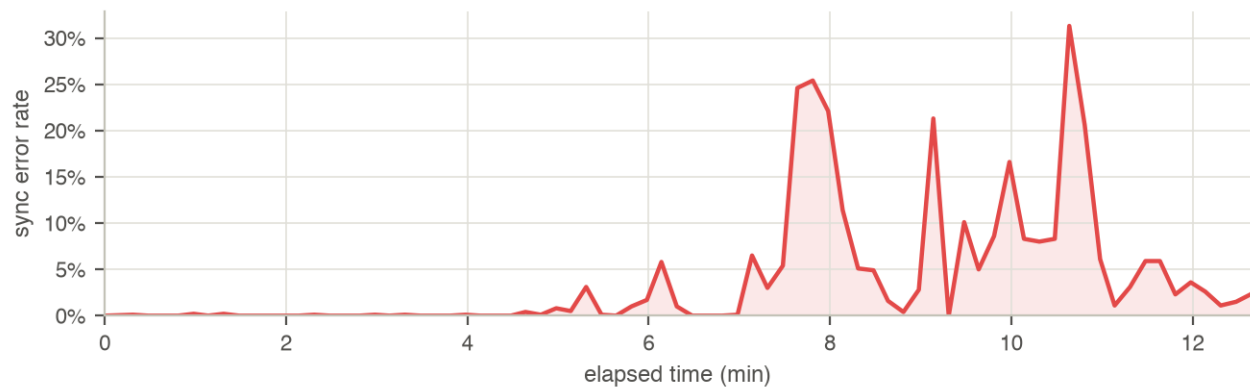
The run's tail (past 12.64 minutes elapsed) is a flat all-zero line - the rover being powered off, not a radio failure - and is excluded from this analysis entirely. All statistics, charts, and conclusions below cover only the first 12.64 minutes of real signal, which this analysis treats as the full 0→2mi run. The minutes-to-miles mapping assumes a steady pace across that whole window - it is a proportional estimate, not a GPS-verified position for each sample.

One more caveat on the numbers themselves: ~400 fixes/s is radio-congestion's synthetic multi-boat load, not real racing traffic - a single real rover runs at roughly 10Hz. Treat the absolute throughput as a stress-test artifact; the range/distance pattern - where the link starts degrading, how it degrades - is the real signal this test exists to capture, and should transfer to real usage.

Fix rate over the run



Sync error rate over the run



Phase summary

Phase	Time / approx. distance	Fix rate avg (min-max)	Error rate avg (max)
1 - strong signal	0.0-4.3 min ~0.00-0.68 mi	397.8 Hz (366.3-402.0)	0.03% (0.2% max)
2 - marginal / bursty	4.5-12.6 min ~0.71-2.00 mi	296.7 Hz (0.0-524.8)	6.03% (31.3% max)

Within phase 2 (50 samples), 13 are severely degraded (fix rate <200Hz or error >10%) while 18 are still near-perfect ($\geq 350\text{Hz}$, <2% error) - the same distance band produces both, sometimes minutes apart.

Transmission-level view (confirmed TX_BATCH_SIZE=4)

The base's own fix_rate_hz already accounts for batching correctly - a batch frame adds its actual fix count (up to 4), not 1, to the fix tally (see baseStation.js's own bandwidthWindow.fixes). But sync_error_rate_hz/error_pct is deliberately measured at the transmission level, not the fix level - a corrupted frame never decodes at all, so there is no per-fix count to attribute the loss to. With TX_BATCH_SIZE=4 confirmed for this run, each lost transmission below actually cost up to 4 fixes at once, not 1 - recomputing directly from the raw frames/sync_errors columns:

Phase	Transmissions/s avg	Fixes/s avg	Total transmissions	Total sync errors	Error rate
1 - strong signal	98.7	394.9	25,674	9	0.04%
2 - marginal / bursty	74.2	296.7	37,095	1,497	3.88%

At batch=4, this radio sent roughly 99 transmissions/second in the clean zone - in the marginal zone, 3.88% of ~37,095 send attempts were corrupted outright (1,497 lost transmissions), each one costing 4 fixes at once rather than 1. Practically: the marginal zone isn't "occasional individual fixes go missing" - it's closer to "roughly 1 in 26 four-fix batches gets wiped entirely." Run unbatched (TX_BATCH_SIZE=1) at the same RF conditions and a similar transmission error rate would likely produce a much gentler-looking fix-rate dip, since each failure would only cost 1 fix instead of 4 - some of this run's dramatic 400-to-60Hz swings are a real RF phenomenon amplified by batching, not purely a property of the RF link itself.

Conclusions

1. No hard range ceiling found within the tested 2 miles. The run's last three real samples (~1.9-2.0mi) show fix rate climbing back to 132→525→522→389 Hz with error down to 1.1-2.3% - the link was recovering, not failing, at the moment the rover was powered off. An earlier pass at this same data (before excluding the power-off tail) concluded the link died by ~1.5mi; that was an artifact of misreading a power-off as an RF failure, not a real finding.

2. The marginal zone starts early - around 0.71mi (t=4.48min) - and never resolves for the rest of the run. From there to 2mi, performance is genuinely unpredictable: the same distance range produces both fully-healthy stretches and severely degraded ones.

3. The degradation pattern is bursty, not a smooth rolloff - and the likely cause is rider/vehicle self-shadowing, not distance or terrain. A true free-space-path-loss story would decline monotonically with distance. This one doesn't - it fully recovers to ~400Hz/~0% error at multiple points inside the marginal zone (t=6.48-6.98min, t=8.64-8.98min, and again right at the end near t=12.31-12.64min) before degrading again. The test rider reports riding an ATV directly away from the base for this run - meaning the rider's own body and the vehicle's mass sat continuously in the direct line-of-sight path between the two antennas, not just at fixed obstruction points along a route. Small changes in the rider's posture, the ATV's exact heading, or bumps in the terrain would shift how much of that mass actually blocks the direct path moment to moment - a highly plausible explanation for exactly this kind of unpredictable, in-and-out pattern, and a more specific cause than generic terrain/Fresnel-zone obstruction.

4. Practical takeaway. Treat ~0.7mi as the point past which you can no longer assume every fix arrives. This data cannot tell you where the link would actually become unusable, since the run ended (on a good stretch) before reaching that point - a longer run, continued until the link fails on its own rather than being switched off, is needed to find the real ceiling.

5. Suggested follow-up test. If self-shadowing is the real cause, repositioning the rover's antenna - mounted higher (clear of the rider's head/shoulders) or off to a side rather than directly behind the body relative to the base - should measurably reduce this choppiness at the same distances. A second run with the antenna relocated, or a run riding perpendicular to the base rather than directly away, would help confirm whether body/vehicle blocking (fixable with mounting position) or genuine terrain/range limits (not fixable that way) is the dominant factor.

Generated from data/base_station_rate_stats_2026-09-27.csv - race-tracker's own LOG_RATE_STATS output.